

Topic 06: Tank Mixing Problems & Draining Rates

Professor Leonard Master Series · Lesson 19

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Giant Kool-Aid Vat

"Think about making a giant vat of Kool-Aid. You have a huge 500-gallon tank. Water with dissolved powder flows in at the top. The mixer stirs it instantly, and the blended mixture drains out of a valve at the bottom."

We want to know: **How many pounds of salt/powder $A(t)$ are in the tank at any minute t ?**

The fundamental law of mass conservation says:

Rate of Change of Salt = (Rate of Salt Entering) – (Rate of Salt Leaving)

$$\frac{dA}{dt} = R_{\text{in}} - R_{\text{out}}$$

Where each rate is:

Rate = (Concentration) $\times$ (Fluid Flow Rate)

$$R_{\text{in}} = c_{\text{in}} \cdot r_{\text{in}}, \quad R_{\text{out}} = c_{\text{out}}(t) \cdot r_{\text{out}}$$

2. Curriculum Reference Diagram

use the fact that $\frac{1}{4}A_0 = A(5730)$ or $\frac{1}{4}A_0 = A_0e^{5730k}$.
 The last equation implies $5730k = \ln \frac{1}{4} = -\ln 2$ and
 so we get $k = -(\ln 2)/5730 = -0.00012097$.
 Therefore $A(t) = A_0e^{-0.00012097t}$. With $A(t) =$
 $0.001A_0$ we have $0.001A_0 = A_0e^{-0.00012097t}$ and
 $-0.00012097t = \ln(0.001) = -\ln 1000$. Thus

$$t = \frac{\ln 2}{0.00002867} \approx 24,180 \text{ years.} \quad \equiv$$

The date found in Example 3 is really at the border of accuracy of this method. The usual carbon-14 technique is limited to about 10 half-lives of the isotope, or roughly 60,000 years. One fundamental reason for this limitation is the relatively short half-life of C-14. There are other problems as well; the chemical analysis needed to obtain an accurate measurement of the remaining C-14 becomes somewhat formidable around the point 0.001A₀.

Figure 6.1: Mixing tank with input and drain streams — from Zill 7th Ed. Chapter 2 (Fig. 2.7.4).

3. Dynamic Volume: The Number One Exam Trap!

What is the concentration leaving the tank, $c_{\text{out}}(t)$?

$$c_{\text{out}}(t) = \frac{\text{Amount of salt in tank}}{\text{Volume of liquid in tank}} = \frac{A(t)}{V(t)}$$

- **Case 1: Constant Volume** ($r_{\text{in}} = r_{\text{out}}$): Liquid enters as fast as it leaves. $V(t) = V_0$.
- **Case 2: Changing Volume** ($r_{\text{in}} \neq r_{\text{out}}$):

$$\text{Net Rate of Liquid Accumulation} = r_{\text{in}} - r_{\text{out}}$$

$$V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t$$

Therefore:

$$\frac{dA}{dt} = c_{\text{in}}r_{\text{in}} - \frac{A(t)}{V_0 + (r_{\text{in}} - r_{\text{out}})t}r_{\text{out}}$$

Rearranging gives a classic First-Order Linear ODE:

$$\frac{dA}{dt} + \left(\frac{r_{\text{out}}}{V_0 + (r_{\text{in}} - r_{\text{out}})t} \right) A = c_{\text{in}}r_{\text{in}}$$

This is solved using the Integrating Factor method from Topic 05!